Designing a Smartwatch-based Micro-Intervention to Support Students' Emotion Regulation

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Emotion regulation is essential for students' well-being and success. However, many students lack emotion regulation skills. This study explores the use of smartwatches to deliver real-time emotion regulation interventions supporting students in stress-full situations. Given their ability to monitor physiological indicators and provide immediate feedback, smartwatches offer a promising solution for delivering emotion regulation micro-interventions. We designed a smartwatch-based micro-intervention and conducted a usability test to evaluate how students interact with the interventions on the smartwatch and their potential to support regulating emotions during stressful situations. Within a controlled laboratory environment, the study involved 13 participants (7 males, 6 females) who were exposed to positive and negative mood induction tasks, followed by emotion regulation interventions delivered via smartwatch notifications. Usability and user experience were assessed using the System Usability Scale (SUS) and the User Experience Questionnaire (UEQ), complemented by qualitative interviews. Our findings provide insights into the potential of wearable technology to support emotion regulation through micro-interventions. However, future research is needed to enhance personalization, usability, and real-world applicability of the proposed smartwatch-based intervention to successfully support emotion regulation.

CCS Concepts: • **Human-centered computing → Usability testing**.

Additional Key Words and Phrases: Emotions, Emotion Regulation, Smartwatch

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1 INTRODUCTION

Emotion regulation plays a pivotal role in mental health and overall well-being [10]. Effective emotion regulation is associated with various positive outcomes, including enhanced performance in various life domains [25, 26]. Students, in particular, encounter unique and intense stressors, ranging from academic pressures to social challenges, making the ability to regulate emotions especially crucial for this population [11]. One of the critical moments when emotion regulation is essential for students is during the job search process [15, 31], particularly at job fairs or during job interviews. These events can be highly stressful, as they often serve as gateways to future career opportunities [24]. The pressure to make a good impression and the competitive nature of job searching can excite significant emotional responses [4].

In this context, wearable technologies present a novel solution for delivering real-time emotion regulation interventions [6]. Smartwatches, in particular, are not only pervasive and easily accessible but also capable of monitoring physiological indicators of stress and providing immediate feedback [7, 17]. Like smartphones, these devices can deliver interventions such as guided breathing exercises, mindfulness prompts, or cognitive reappraisal techniques [3, 23], but on the other side, smartwatches are more subtle. The potential benefits of using smartwatches for emotion regulation are significant, especially in high-stress environments [7]. Successfully regulating emotions in such settings could improve students' performance, confidence, and overall experience. Furthermore, the real-time nature of smartwatch interventions aligns well with the immediate needs of emotion regulation, providing support exactly when students need it.

This study explores a smartwatch-based micro-intervention's usability and user experience for emotion regulation in a controlled environment. By conducting a usability test, we assess how students interact with emotion regulation interventions and evaluate their impact on managing emotions during stressful scenarios. Our research provides valuable insights into the practical applications of wearable technology for emotion regulation and its potential to enhance student outcomes in high-stress situations. By investigating the interventions in a controlled environment as a first step, we lay the foundation to derive effective strategies to support students in managing their emotions, ultimately contributing to their well-being and improved performance. Therefore, we articulate the following research question (RQ): *How to design smartwatch-based micro-interventions to support students' emotion regulation in stressful situations?*

With our research, we aim to contribute to the Human-Computer Interaction (HCI) field by suggesting a design for smartwatch-based micro-interventions to support emotion regulation. We believe that smartwatches represent a promising technology to deliver micro-interventions to support emotion regulation for students in stressful situations.

2 RELATED WORK

Our research is related to prior literature on emotion regulation, smartwatch technology, and micro-interventions. Emotion regulation involves the extrinsic and intrinsic processes by which individuals influence which emotions they have, when they have them, and how they experience and express these emotions [9, 30]. Therefore, emotion regulation is a critical aspect of psychological research, as it can influence individuals' well-being and performance [25, 26]. The effectiveness of emotion regulation strategies can greatly depend on the context in which they are applied [1, 2]. Effective emotion regulation strategies can be linked to different positive outcomes, including reduced anxiety and depression [28], enhanced academic performance [12], and improved interpersonal relationships [26].

Smartwatch technology has advanced rapidly in recent years, offering a range of features that can be leveraged for health and well-being interventions [14, 32]. Smartwatches have sensors that can monitor physiological indicators such as heart rate, skin conductance, and accelerometer data, providing valuable real-time data on the user's state [17]. This capability makes them an ideal platform for delivering interventions to help manage stress and emotions [14]. Lima et al. [21] highlights the potential of wearable technology, including smartwatches, to support health-related behavior changes through continuous monitoring and feedback. As these devices are lightweight and easily integrated into daily life, smartwatches can be suitable for delivering different types of interventions.

Micro-interventions refer to brief, targeted strategies designed to address specific issues or promote particular outcomes. These interventions are designed to be quick and easy to implement, making them suitable for integration into daily routines [29]. Micro-interventions have predominantly been tested and implemented using smartphones as the primary technological device [3, 22, 23]. In the context of emotion regulation, interventions can include techniques such as breathing exercises or mindfulness prompts [13]. Different research has shown that even brief interventions can positively affect emotional well-being and stress management [5, 16, 18].

3 RESEARCH METHOD

In the following we explain the research process we followed to design and evaluate smartwatch-based interventions for emotion regulation. We provide a short description of our initial problem awareness study (1), followed by the evaluation study design (2), and finally we present the proposed design of the smartwatch-app (3).

3.1 Problem Awareness Study at Job Fairs

To explore the problem of emotion regulation in stressful situations, we collected data at three different job fairs with 36 subjects to better understand relevant emotions during a job fair as a starting point to provide emotion regulation interventions. Participants of the job fair wore a smartwatch provided by us. The smartwatch included a basic experience sampling app that asked around every 10 to 15 minutes the students about their major emotions (relaxed, happy, sad or angry) at that specific moment. In addition to the emotion reports, we collected biosignals (e.g., heart beat rate, resting heart rate, or heart rate variability). After the study participation, we asked the subjects about their experiences and, in particular, what "emotional" challenges they faced during their job fair visit.

Overall, we collected 226 emotion reports from 36 different subjects. Around 50.0 % of the reports are “relaxed”, around 40.0 % of the reports are “happy”, 6.0 % of the reports are ”angry” and finally 4.0 % are the report “sad”. At every job fair the report “relaxed” was the most used. Additionally, we collected 220 context reports of the 36 different subjects. The most reported context was “other” with 35.0 %, next the context “conversation” with 32.0 %, followed by the report “looking at the booth” with 19.0%, and finally the report “walking” with around 14.0 %. To connect both emotion and context reports, “happy” and “other” were reported most often, followed by “happy” and “conversation”. For the negative emotions, the most reported connection were “angry” with “conversation” and “sad” with “other”. Following the results, we can see that negative emotions, specifically sadness, and anger, build 10.0 % of the collected reports. Looking only at the negative emotions, the context of conversations is reported the most, highlighting the conversations at job fairs as emotionally challenging. These insights on the emotional landscape during job seekers form the basis for conceptualizing the intervention design to support job seekers in this specific situation.

3.2 Usability Test of the Smartwatch-based Micro-Intervention

The usability test was conducted in an experimental laboratory to ensure a consistent and distraction-free environment for all participants. 13 participants were recruited. 6 of the participants were female and 7 male. Regarding the age range of the participants, 2 participants were 18-20 years old, 7 were 21-25 years old, 3 were 26-29 years old and finally one participant was 30-39 years old. An overview of the participants is shown in Table 1.

Sessions took about 60 minutes and the participants received a fixed participation fee of €11.00 plus a bonus up to €4.00 depending on their performance in the math tasks (overall mean payment: €13.00). We created a within-subjects experimental design with two different task orders (first easy task followed by difficult task (n=7)/ first difficult task followed by easy task (n=6).

Each participant completed the usability test individually, allowing for precise monitoring and data collection. The usability test proceeded as follows:

(1) At the beginning, participants receive a thorough introduction to the study and an explanation of the procedure. Participants are trained on how to use the smartwatch and respond to and interact with notifications.
(2) The test is divided into two main blocks: one designed to induce a positive emotion and the other to induce a negative emotion. In the positive emotion block, participants solve relatively easy math problems designed to

Table 1. Participants of the Usability Test

Participant	Gender	Age Group
P1	male	21-25
P2	female	21-25
P3	male	21-25
P4	female	21-25
P5	male	26-29
P6	female	21-25
P7	male	21-25
P8	female	26-29
P9	male	18-20
P10	female	30-39
P11	male	21-25
P12	female	26-29
P13	male	18-20

build confidence and induce a positive emotional state. Following the math tasks, participants watch a series of uplifting and positive videos [27]. In the negative emotion block, participants then move on to solving a set of challenging math problems under time pressure intended to induce frustration and negative emotional states [19]. After the math tasks, participants watch distressing and negative videos to reinforce the negative emotional states [27]. After completing each block, participants take a short break, receiving a notification from the smartwatch prompting them to engage in the micro-intervention supporting emotion regulation. These interventions include a request to identify the participants' emotion followed by a motivational feedback for the positive stream and a short breathing exercises followed by a motivational feedback for the negative stream, to tailor the emotion they were in (see Fig. 1).

(3) Following the completion of both emotion blocks and interventions, participants fill out two standardized questionnaires: the System Usability Scale (SUS) [8] to assess the overall usability of the smartwatch intervention system, and the User Experience Questionnaire (UEQ) [20] to evaluate their experience and satisfaction with the smartwatch-based interventions. Additionally, participants take part in a short semi-structured interview to provide deeper insights into their experiences, perceptions of the interventions, and any suggestions for improvement. Example questions during the interview are *"What kind of notifications did you get from the smartwatch?"*, *"Did the notification on the smartwatch match your emotional state in the respective situation? Why or why not?"*, or *"Do you see additional ways in which a notification on the smartwatch could support your emotional state? What would that look like?"*. The interview is recorded upon consent given by the participants.

The test aims to evaluate the relevance and applicability of smartwatch-based interventions supporting emotion regulation by guiding participants through this structured process and collecting detailed feedback.

3.3 Implementation of the Smartwatch App

We developed an Android app using Android Studio and the Kotlin programming language to implement smartwatch notifications. This app was deployed on a Samsung Watch 5 Pro. For this, we created a view for each part of the study and implemented a state manager that saved the participant's current process. When the participant clicked that a part of the study was finished, a worker was started, which triggered the notification for the intervention in 30 seconds.

For the notifications, we built a periodic worker using the Androidx work package[1]. This worker used the Android Notification Manager[2] to notify the participant. The notification was passed to the system with the highest priority to ensure a fast execution by the system. This notification was displayed on the clock's main screen (watch face) or in the notification feed for 3 minutes. With one click, the participant can then go to the report screen. The app asks participants about their emotional state, divided in either positive - "happy or relaxed" or negative - "sad or angry" followed by the related intervention (see Figure 1). The implemented notifications are based on the emotion regulation process described by Slovak et al. [29]. First the identification and selection is implemented through the question and necessary confirmation or refusal of the expected emotion in the respective situation. For the following interventions we chose an experiential and on-the-spot delivery mechanism through the smartwatch notifications. For both, positive and negative emotion connected interventions, we relied on response modulation through reflection support and in addition for the negative connected intervention on a short breathing exercise. If the participant does not choose any of the emotional states, the following message is shown: "Thanks for your answer".

Fig. 1. Smartwatch-based Interventions for Emotion Regulation

4 RESULTS

The usability test included 13 participants. Overall, five of the participants use a smartwatch in their daily lives. Four of the participants selected both emotion interventions (positive and negative) and also consciously perceived them, three participants stated that they only had positive emotions and therefore only perceived the positive intervention. Six participants perceived the prompts to confirm or reject the suggested positive or negative emotions, but could not remember the following intervention screens on the smartwatch.

[1] https://developer.android.com/jetpack/androidx/releases/work
[2] https://developer.android.com/reference/android/app/NotificationManager

The SUS was used to assess the overall usability of the smartwatch intervention system. The average SUS score was 81.67 %, indicating a generally acceptable level of usability. This score suggests that while the system is usable, there is room for improvement to enhance user satisfaction and ease of use. The UEQ provided further insights into participants' user experience with the smartwatch interventions. The UEQ results, shown in Figure 2, indicated that participants found the interventions moderately engaging and helpful but also underlined several areas where improvements could be made to better meet user needs. Best rated aspects are connected to perspicuity (understandable, easy to learn, easy, clear) and efficiency (organized, practical) of the smartwatch notifications. In comparison, the aspects regarding novelty (conventional, usual, conservative) were rated rather poorly. For the SUS, all 13 participants answered the questionnaire,

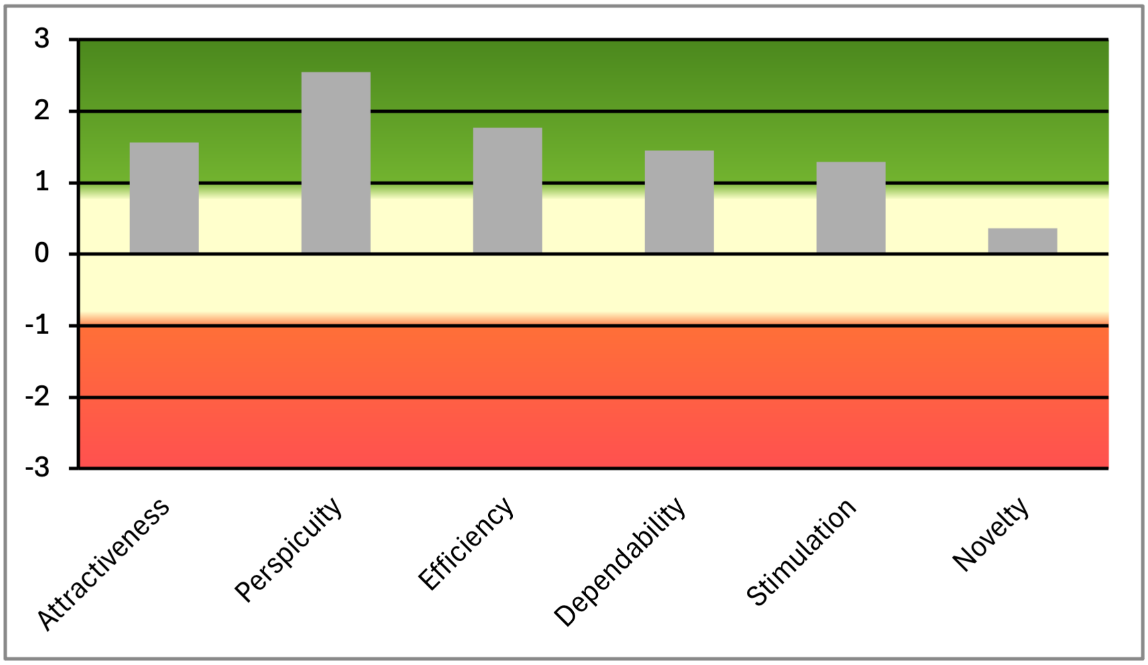

Fig. 2. Results of the User Experience Questionnaire

while 11 did so for the UEQ due to a technical issue when recording responses.

Additionally, participants provided valuable qualitative feedback on various aspects of the interventions. First, the intervention following the negative emotion block was positively received, *"Taking a short break and breathing deeply really helped me to calm down"* (P11). Besides, one participant (P4) mentioned that the *"display of animations could further support my motivation"*, indicating that visual enhancements might increase the effectiveness of the interventions. Another (P5) suggested, *"I could imagine that breathing or meditation exercises would support me. [...] I would also need variety, so not always the same message."*. Suggestions for adding *"smileys and reminders to take a break or drink something"* (P12), underline the need for more variation in the visualisation of possible interventions. Some participants expressed concerns about notification frequency, *"There definitely shouldn't be too many notifications, or I'd quickly feel disturbed."* (P7). This was connected with the issue of maintaining focus, as *"the messages should definitely not disrupt my focus"* (P6). Lastly, a desire for more authentic feedback was expressed, *"I would like realistic feedback, not just 'well done,' but also 'yes, that wasn't so good.'"* (P13).

5 DISCUSSION

The usability and user experience survey data, combined with the qualitative data collected in the interviews, enable a better understanding of how students perceive the smartwatch-based emotion regulation interventions. The SUS score indicates a satisfactory level of usability, though the qualitative feedback highlights specific areas for enhancement. Regarding the UEQ results, participants find the product overall very clear and easy to understand, as perspicuity receives the highest positive score. Additionally, participants evaluate the efficiency very positively, meaning they can complete their tasks quickly and effectively. The design seems to be intuitive and user-friendly, but less innovative, as the participants rate novelty with the lowest score. Overall, participants appreciated the potential of the interventions to support emotion regulation but also raised concerns about the frequency of notifications, the authenticity of content, and privacy issues. These insights will be critical for refining the intervention strategies to better support emotion regulation and user engagement, ultimately contributing to developing more effective tools for managing emotions in high-stress environments.

The study involved only 13 participants, which limits the generalizability of the results. A larger and more diverse sample would provide more robust data and potentially more varied insights into the usability and effectiveness of the smartwatch interventions. The study was conducted in a controlled laboratory setting, which may not accurately reflect real-world conditions. Participants' responses and interactions with the smartwatch interventions might differ in naturalistic settings, such as job fairs or other high-stress environments. The study focused on a limited range of interventions (e.g., breathing and reflection support). Exploring a broader array of emotion regulation strategies and their integration into smartwatch technology would provide a more comprehensive understanding of what works best for different users. Objective measures, such as physiological data collected by the smartwatch, could complement self-reported data and provide a more complete picture of the interventions' effectiveness. The study induced only positive and negative moods through specific tasks and videos. Emotion regulation involves a wide spectrum of emotional states, and future research should consider a broader range of emotions and contexts to better understand the versatility of smartwatch interventions.

The findings from this study provide valuable insights into the potential of smartwatch technology for emotion regulation. The positive reception of the interventions' potential, combined with the constructive feedback on usability and content, underscores the importance of iterative design and human-centered development. Future research will focus on enhancing the dynamic nature of the interventions, ensuring they are varied and engaging to maintain user interest. Additionally, careful consideration must be given to the balance between helpful notifications and user privacy, ensuring that interventions are supportive without being intrusive. The qualitative feedback also points to the necessity of realistic and personalized feedback within the interventions. Incorporating user preferences and providing varied motivational content can help in making the interventions more effective and credible.

6 CONCLUSION

This study explored the feasibility and effectiveness of using smartwatch-based interventions for emotion regulation among students in high-stress situations, such as job fairs. The usability test conducted in a controlled laboratory environment involved 13 participants. The participants were guided through two emotion induction blocks, one designed to induce positive emotions and the other negative emotions, using a combination of math tasks and emotional videos. The study assessed the usability and user experience of the proposed smartwatch interventions through the SUS and UEQ, supplemented by qualitative interviews. The usability test highlights the potential of smartwatch-based interventions to support emotion regulation in students during high-stress scenarios. While the initial usability and user experience results are promising, the qualitative feedback emphasizes the need to further refine the interventions. By addressing these concerns and building on the identified positive aspects, future refinements can create more effective, user-friendly smartwatch interventions for managing emotions, contributing to better performance and well-being for students in challenging environments.

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